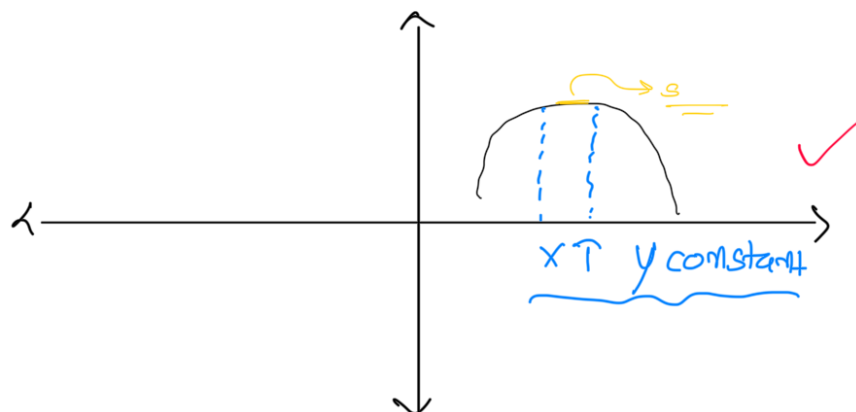
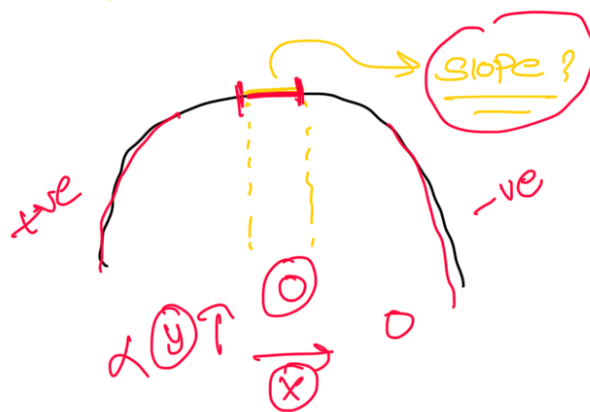
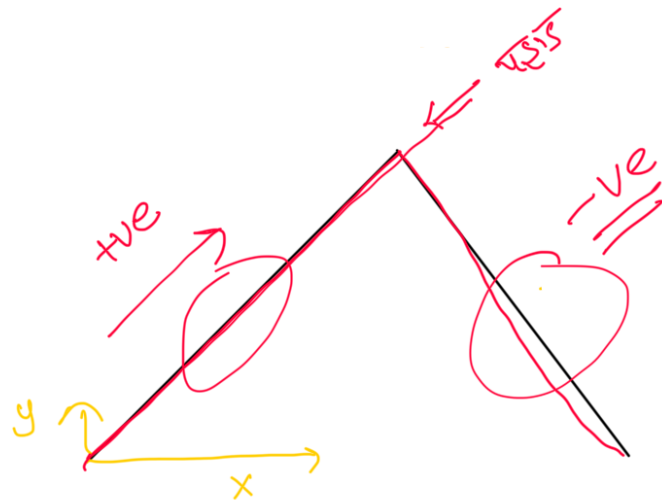
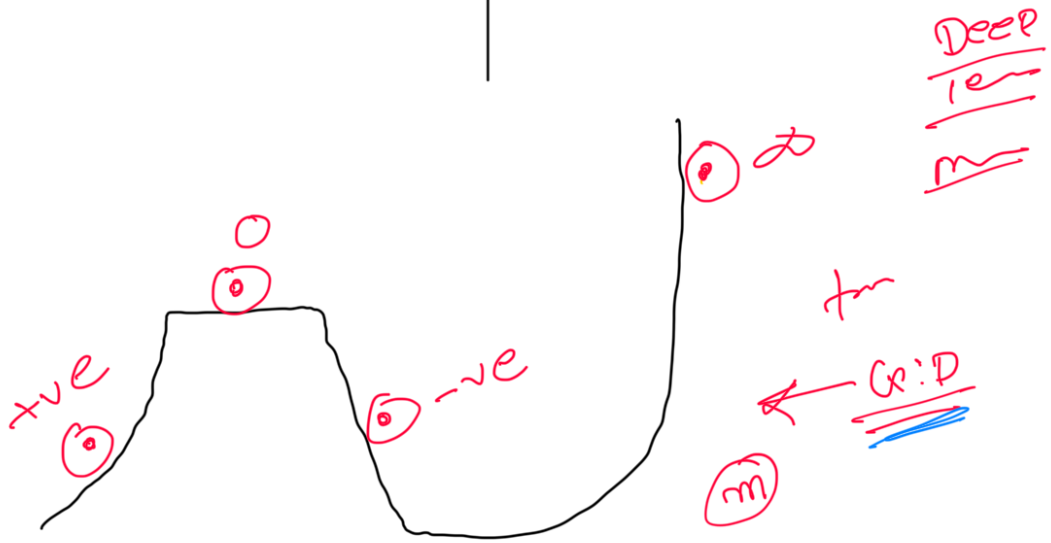
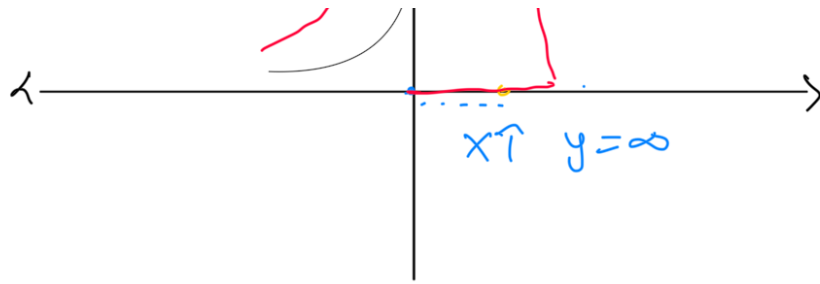


* Revision:

① Slope





→ and how we measure slope: tano

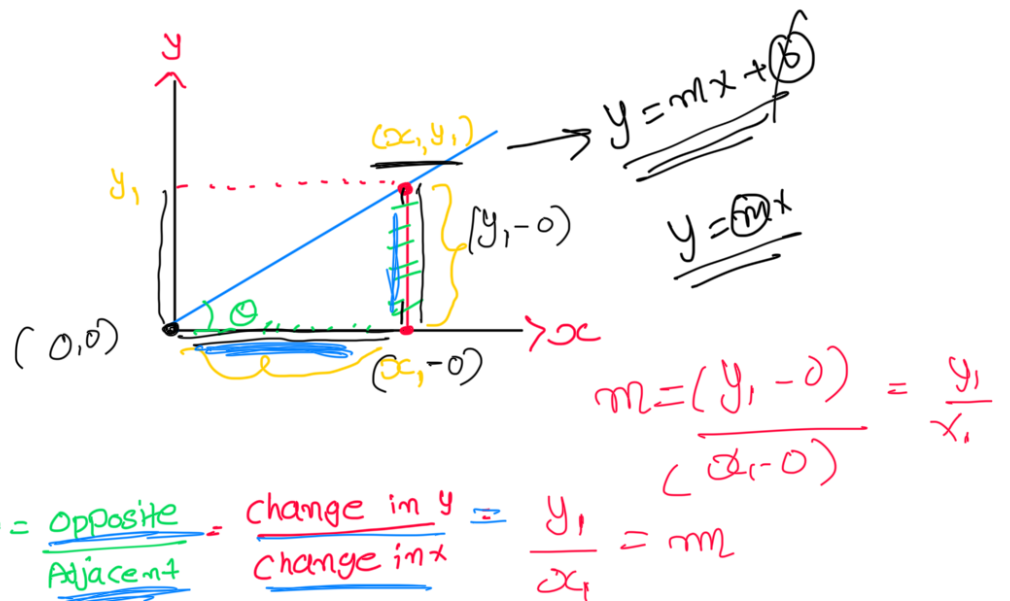
↳ 2nd way we will see it later is

$$\frac{dy}{dx}$$

↓ definition

Slope describes how much one variable changes in response to a change in another variable.

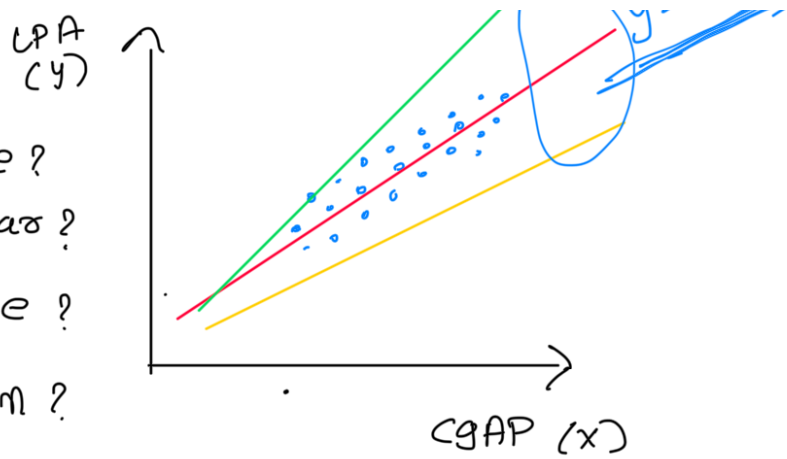
$y = 2x$
 $m(\text{slope})$
 $\frac{y}{x} = 2$



② Simple Linear Regression:

$y = mx + b$

Why it is simple?
 Why it is linear?
 Best prediction line?
 Line eqn equation?



- ① bcz there is only one x/p feature.
- ② bcz it can draw only Line (straight)
- ③ minimum Error.
- ④ $y = mx + b$

③ Error & Regression Matrix:

a: MSE

b: RMSE

c: MAE

d: R^2

e: Adj R^2

Revision

we will see today.

$$\frac{(y - \hat{y})^2}{n}$$

$$\frac{|y - \hat{y}|}{n}$$

④ Outliers: data point that is significantly different from the rest of the data.

→ effect of it: Pull the model from the actual trend.

observablehq.com

→ how to detect:
IGR
Z score
scatter plot
Box plot

} we will see this in statistics class.

→ is it useful in any cases?

↳ think about it we will answer it in end of the class.

⇒ Lecture: 5

Sim!: 1 fea (x)

① Multiple linear Regression:

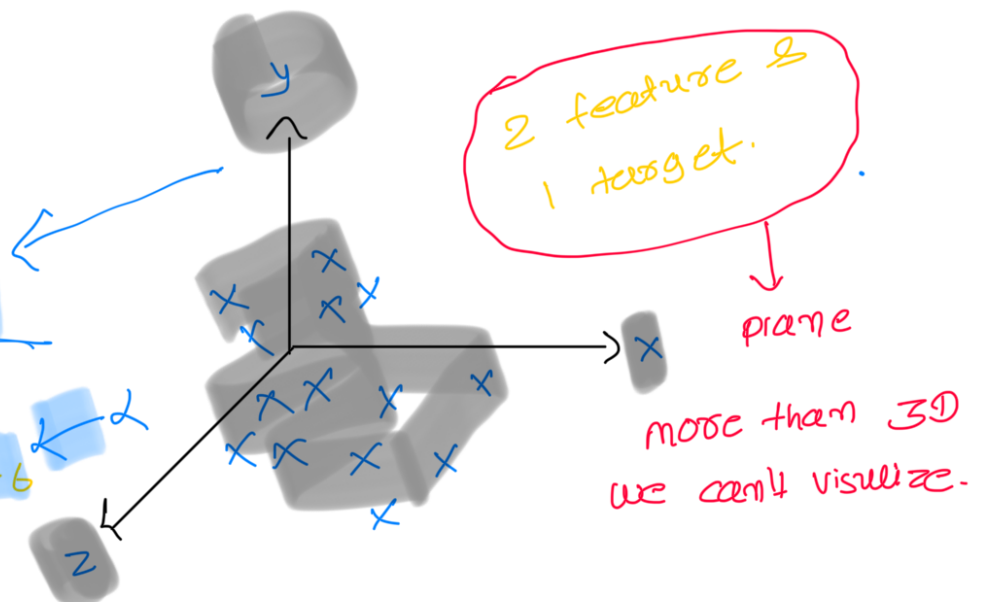
IQ | CGPA | 12th per. | 10th per | LPA
x y
(5 features)

1 feature:
 $y = mx + b$

2 fea:
 $y = m_1x_1 + m_2x_2 + b$

...

$y = m_1x_1 + m_2x_2 + \dots + b$



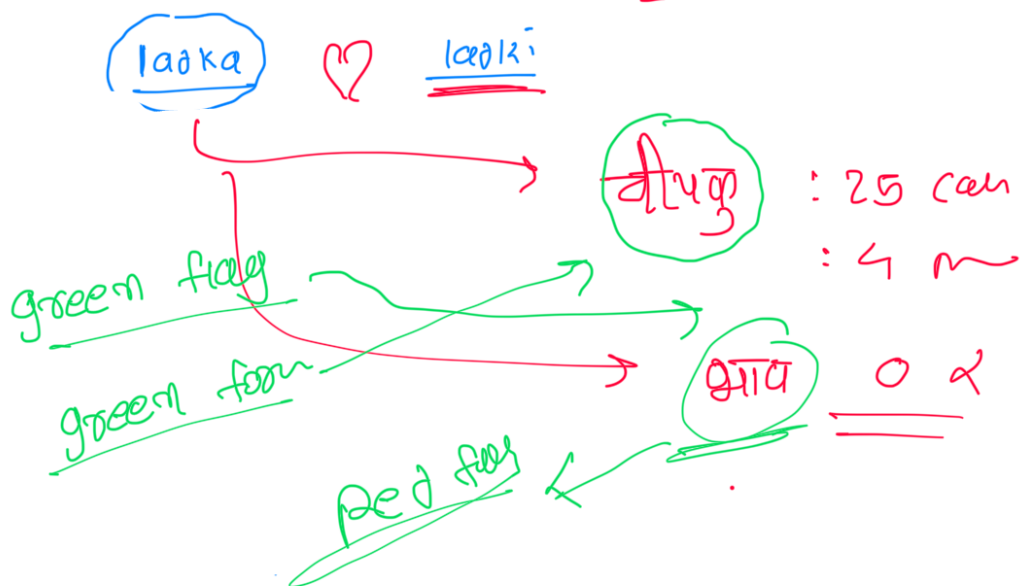
multiple-regression-visualizer.onrender.com

Same as LR (only no of features are high.)

② Overfitting & underfitting:

over: ज्यादा

under: कम

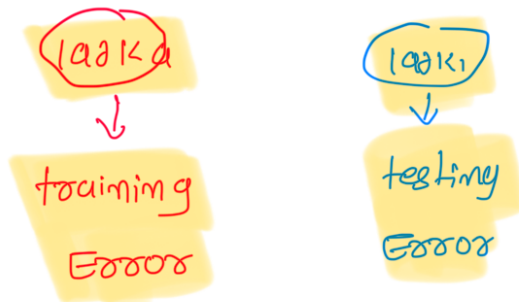


laaka	laaki	Relationship
<u>green form.</u>	<u>Pasam hai n</u>	✓ → Per
<u>red form</u>	<u>Pasam hai n</u>	α } → <u>Under</u>
<u>green</u>	no	α } → <u>overfitting</u>
<u>Red</u>	<u>no</u>	✓ → Per

Rule of Relationship (healthy):

laadki ke jese pasand hai
Aese Rahan hai.

model should capture the behavior of the
Data.

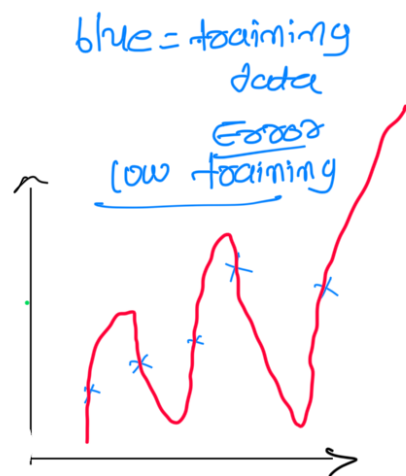
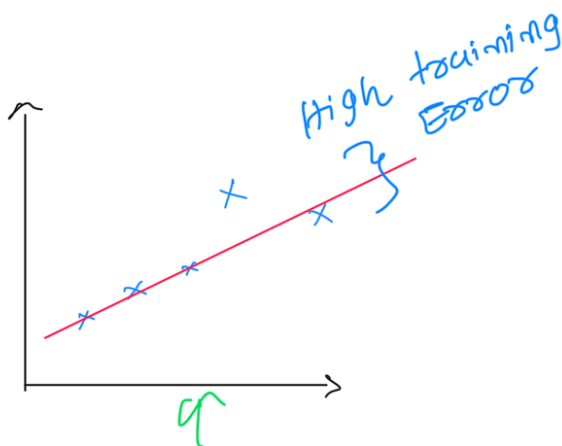


training Error $\approx$ testing Error

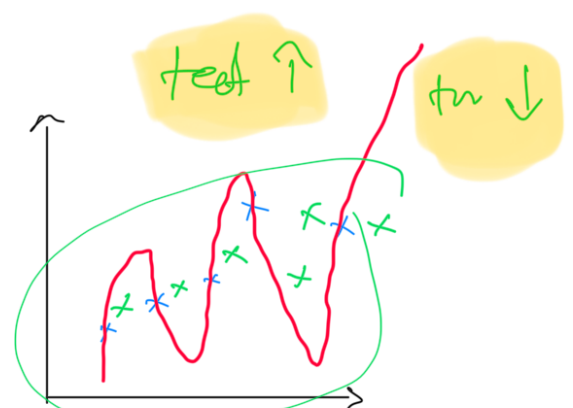
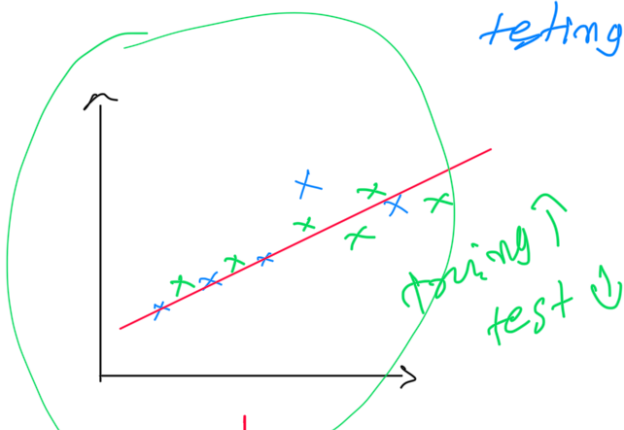
training Error $\ll$ || : $\uparrow$ $\rightarrow$ over fit

training Error $\uparrow$ & testing Error $\uparrow$ $\rightarrow$ 0-mo

training Error $\downarrow$ & || || $\downarrow$: perfect



We will NOT Answer without
testing data.

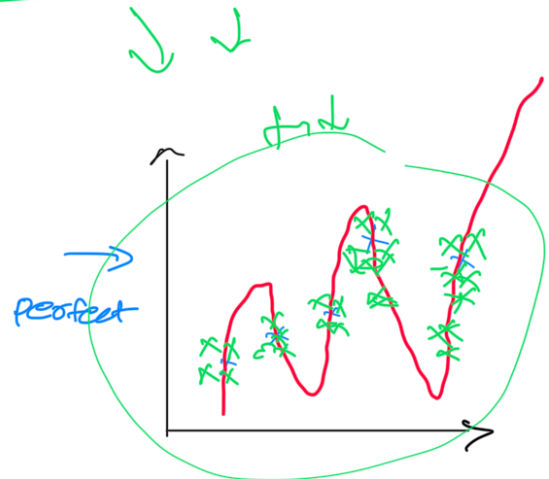
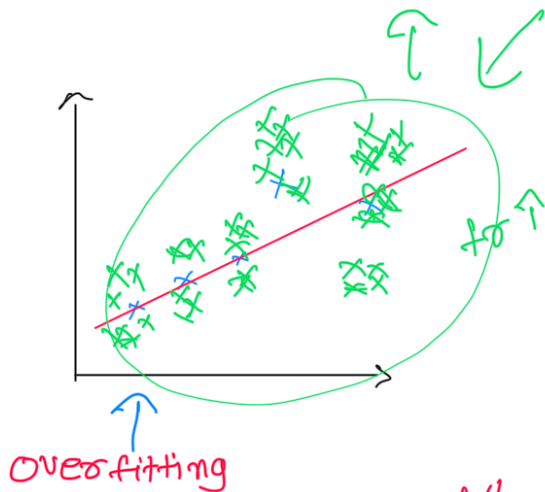


↓
perfect. both low.

↑
overfitting

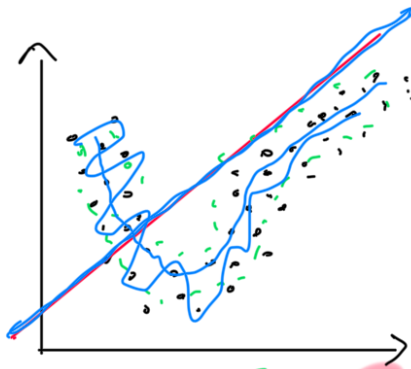
training errors low
 testing " high.

Testing

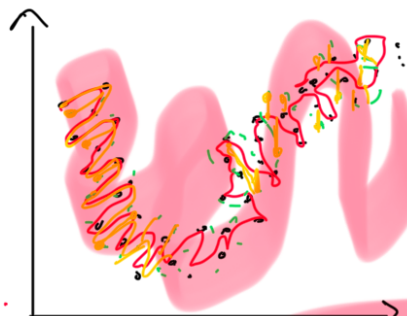


training errors ↓
 testing errors ↑

Why over/under depends on
testing data. or we can say
behaviour of data.



2nd

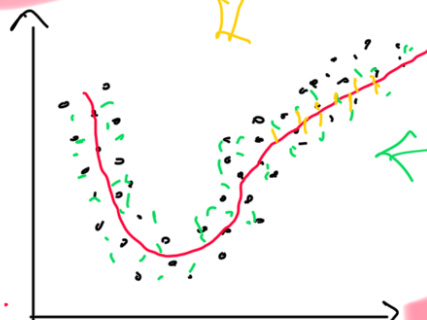


green:
test
 black:
training

over
 training ↑
 testing ↑

perfect test ↑
 train lowest
 test highest

3rd
 high



Underfitting.

test middle

both ↓

Underfitting: model is too simple. : $tr \uparrow$ $test \uparrow$

overfitting: noisy / memorizes : $tr \downarrow$ $test \uparrow$

good model: perfect spot between : train ↓ testing ↓
complex & simple. middle

* When overfitting:

- for simple task : complex model
- Training data is too small.
- Too many features.

* When underfitting:

- Simple model
- Too few features.
- Excessive Regularization

Q: When outliers are helpful or important ?

- Fraud detection
- Anomaly "
- Rare events